VENKATRAMANAN MAHENDRARAJAN

Scientist (ICMR Project)

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Research Metrics Citations: 203, h-index: 4, i10-index: 4

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https://venkatramananmahendrarajan.github.io

ABOUT ME

I am a dedicated researcher with expertise in microbiology, molecular biology, bioinformatics and animal handling for preclinical studies. My work spans bacterial membrane vesicles, antimicrobial peptides, gene expression analysis, and computational drug discovery. I have extensive experience in sequencing data analysis for Illumina and Nanopore platforms, as well as transcriptomics-based differential gene expression analysis. Passionate about translational research, I aim to bridge computational and experimental biology for innovative solutions in biotechnology and healthcare.

EDUCATION

Integrated MSc Life Sciences

Central University of Tamil Nadu

2015-2020

CGPA 8.3/10, First class with Distinction

SKILLS

Microbial techniques
Molecular techniques
Animal Handling
Computational - Docking & Dynamics
Sequencing - Nanopore & Illumina
Microscopy - CLSM & SEM
PCR - Conventional & q RT PCR
Flow cytometry
Proteomics - SDS PAGE & Western blot

PROFESSIONAL EXPERIENCE

2024 – Present
ICMR Project Scientist (Doctoral Research Fellow)
Vellore Institute of Technology, Vellore

2022 – Present
Doctoral Research Fellow
Vellore Institute of Technology, Vellore

2021 – 2022
SERB Junior Research Fellow
Central University of Punjab, Punjab

EXAM QUALIFIED

Gradual Aptitude Test for Engineering (GATE) - Life Sciences, 2020, India.
Percentile - 78.6

TEACHING EXPERIENCE

2022 – 2023
Teaching cum Research Assistant
Vellore Institute of Technology

PUBLICATIONS

[1] **Venkatramanan M,** Sankar Ganesh P, Senthil R, Akshay J, Veera Ravi A, Langeswaran K, et al. Inhibition of Quorum Sensing and Biofilm Formation in Chromobacterium violaceum by Fruit Extracts of Passiflora edulis. ACS Omega 2020;5:25605–16. https://doi.org/10.1021/acsomega.0c02483. IF: 4.3

[2] **Mahendrarajan V**, Bari VK. A critical role of farnesol in the modulation of Amphotericin B and Aureobasidin A antifungal drug susceptibility. Mycology 2022;13:305–17. https://doi.org/10.1080/21501203.2022.2138599. IF: 4.4

[3] **Venkatramanan M**, Nalini E. Regulation of virulence in Chromobacterium violaceum and strategies to combat it. Front Microbiol 2024;15:1–13. https://doi.org/10.3389/fmicb.2024.1303595. IF: 5.2

[4] **Mahendrarajan V**, Lazarus H, Easwaran N. Quorum quenching mediated biofilm impediment in Chromobacterium violaceum and Staphylococcus aureus by leaf extracts of Delonix elata. Heliyon 2024;10:e31898. https://doi.org/10.1016/j.heliyon.2024.e31898. IF: 4

[5] **Mahendrarajan V**, Easwaran N. Quorum quenching effects of linoleic and stearic acids on outer membrane vesicle-mediated virulence in Chromobacterium violaceum. Biofouling 2024:1–11. https://doi.org/10.1080/08927014.2024.2446930. IF: 2.6

[6] **Mahendrarajan V**, Nalini E. In silico analysis of bacteriocins from Lactobacillus acidophilus membrane vesicles against Streptococcus mutans GtfB protein. J Biomol Struct Dyn 2025;0:1–11. https://doi.org/10.1080/07391102.2025.2460743. IF: 2.7

LANGUAGE

- Tamil (Native)
- English (Proficient)
- Hindi (Intermediate)

PROFESSIONAL MEMBERSHIP

2024 – Present | Contributing Membership
American Society of Microbiology

2025 – Present | EXO-RISING Membership
Indian Extracellular Vesicles Society

SCIENTIFIC PRESENTATIONS

2022 | Poster Presentation
IMMUNOCON- Annual Conference of the Indian Immunology Society, PIGMER, Chandigarh.

2023 | Poster Presentation
International Conference on Biotechnology for Environment & Health (ICBEH), VIT, Vellore.

2024 | Poster Presentation
International Conference on Contemporary Antimicrobial Research (ICCAR), NEHU, Shillong.

RESEARCH TRAININGS

2024 | Extracellular Vesicles: Isolation, Characterization & Applications, AMRF, Madurai

2025 | Training on the state of art advanced multi omics techniques in Molecular Oncology, ICMR, Govt. of India, Cancer Institute (WIA), Chennai.

ONGOING RESEARCH WORK

Parkinson's Disease Multi-Omics Integration
Conducted integrative multi-omics analysis combining brain transcriptomics and gut microbiome metagenomics datasets from control and Parkinson's disease cohorts to investigate the gut–brain axis. Performed differential gene expression analysis using DESeq2 and carried out comprehensive microbiome profiling using HUMAnN and MetaPhlAn. Mapped host–microbiome interactions to identify functionally relevant associations contributing to neuroinflammation. Applied machine learning approaches, including Random Forest-based feature selection, to identify key predictive signatures and potential biomarkers associated with Parkinson's disease pathology

[7] **Mahendrarajan, V.,** Kodiveri Muthukaliannan, G., & Easwaran, N. (2025). Quorum quenching mediated virulence factor impairment by Pandanus tectorius leaf extract against Chromobacterium violaceum . Natural Product Research, 1–8. https://doi.org/10.1080/14786419.2025.2491837. IF: 2

[8] **Venkatramanan, M** & Easwaran, N. (2025). Isolation, Probiotic Characterization and Genomic Analysis of Enterococcus durans VIT3 from Edible Curd. Microbial Pathogenesis. 107649. 10.1016/j.micpath.2025.107649. IF: 3.3

[9] **Mahendrarajan V**, Lazarus HPS, Muthukaliannan GK, Varghese S, Easwaran N. Membrane vesicles from Red Complex bacteria: key players in oral pathogenesis, immune disruption, systemic diseases, and therapeutic insights. Front Oral Health. 2025 Jul 28;6:1607931. doi: 10.3389/froh.2025.1607931. IF: 3.1

[10] **Mahendrarajan, V.** and Easwaran, N. (2026) 'Integrated proteomics, molecular dynamics, and in vitro characterization of antimicrobial peptide from Lactobacillus acidophilus vesicles against Streptococcus mutans', RSC Adv., 16(7), 5743–5757. https://doi.org/10.1039/D5RA09511E. IF: 4.6

[11] Lazarus, H. P. S., **Mahendrarajan, V.**, & Easwaran, N. (2026). Genomic insights into Exiguobacterium indicum VIT-2023: a dual-function PGPR with potential for plant health protection and plastic biodegradation. Environmental Pollutants and Bioavailability, 38(1). https://doi.org/10.1080/26395940.2026.2636356. IF: 3.2

[12] **V. Mahendrarajan**, N. Easwaran, Biocompatible Membrane Vesicles from Lactobacillus acidophilus MTCC 10307 Exhibit Potent Anti-Inflammatory Activity, BioRxiv. (2026). https://doi.org/10.64898/2026.04.01.715785.

[13] **Mahendrarajan V** et al. (2026). Lactobacillus acidophilus membrane vesicles demonstrate antimicrobial and antibiofilm activity against Streptococcus mutans – an in vitro and in vivo study. Artificial Cells, Nanomedicine, and Biotechnology, IF 4.5 (Accepted)

[14] **Mahendrarajan V** et al. (2026). Triphala Modulates Membrane Vesicle Transcriptome of Enterococcus durans VIT3 to Influence Antibiotic Response and Probiotic Functions. Molecular biotechnology, IF 2.5 (Accepted)

PATENT PUBLISHED

[1] Easwaran, N., **Mahendrarajan, V.,** and Lazarus, H.P.S., A process for extraction and purification of outer membrane vesicles from complex biofilm architectures of Chromobacterium violaceum. Application No.202541131837 A

PATENT FILED

[1] Easwaran, N. and **Mahendrarajan, V.,** A synthetic anticacterial peptide VMNE244 derived from Lactobacillus acidophilus membrane vesicles and its use against Streptococcus mutans

REFERENCES

Dr. Nalini. E - Vellore Institute of Technology, Assistant professor
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Dr. Gothandam. K. M - Vellore Institute of Technology, Professor
Email: k.m.gothandam@vit.ac.in
Dr. Shankar. E. M - Central University of Tamil Nadu, Professor
Email: shankarem@cutn.ac.in